Comparison experiment on gpt-4o

Number of samples: 224 (85 from TruthfulQA, 63 from TriviaQA, 76 from HotpotQA)

Experiment result parameters:

Total hallucination: hallucination in **base_response**

Total hallucination after repair: hallucination in **final_answer** provided by approaches

Hallucination repair: **base_response** is hallucination, **final_answer** is not hallucination

unsuccessful repair: **base_response** is hallucination, **base_response** != **final_answer**, **final_answer** is hallucination

Unable repair: **base_response** is hallucination, **base_response** == **final_answer**

Unnecessary repair: **base_response** is not hallucination, **final_answer** is hallucination

Avg token cost pre question

Avg time cost pre question

Experiment results:

Our approach:

Total hallucination: 49/224 (21.88%), total hallucination after repair: 32/224 (14.29%)
Hallucination repair: 22/49 (44.90%), unsuccessful repair: 16/49 (32.65%), unable repair: 11/49 (22.45%)
Unnecessary repair: 6 (2.68%)
Avg token cost pre question: 508.42
Avg time cost pre question: 2.2850s

ProCo-single iteration

Total hallucination: 66/224 (29.46%), total hallucination after repair: 74/224 (33.04%)
Hallucination repair: 17/66 (25.76%), unsuccessful repair: 18/66(27.27%), unable repair: 31/66(46.97%)
Unnecessary repair: 25
Avg token cost pre question: 2453.25
Avg time cost pre question: 20.5772s

ProCo-3 iterations

Total hallucination: 66/224 (29.46%), total hallucination after repair: 65/224 (29.02%)
Hallucination repair: 17/66 (25.76%), unsuccessful repair: 20/66 (30.30%), unable repair: 29/66 (43.94%)
Unnecessary repair: 16
Avg token cost pre question: 3781.87
Avg time cost pre question: 30.087s

Our experiment result on GPT-5 with auto-thinking:

Total hallucination: 16/224 (7.14%), total hallucination after repair: 11/224 (4.91%)
Hallucination repair: 7/16 (44.90%), unsuccessful repair: 6/16 (37.50%), unable repair: 3/16 (18.75%)
Unnecessary repair: 3 (1.34%)
Avg token cost pre question:-
Avg time cost pre question:-

Evaluation:

Idea:

1. ProCo(from paper: Large Language Models Can Self-Correct with Key Condition Verificatio, **emnlp2024**) did not perform well in general QA process, although it shows great performance in math and choice-making QA, even in their paper shows that they only have 47% exact match on HotpotQA and NQ(Natural Questions).
2. Source code of CoVe(from paper:Chain-of-Verification Reduces Hallucination in Large Language Models, **META AI**) has been deployed, we can try to run this as another baseline
3. ProgCo(from paper: ProgCo: Program Helps Self-Correction of Large Language Models, **ACL 2025**) we should also try this method, however their codes are hard to read and modify.
4. Run comparison experiment on gpt-5 without auto-thinking